# FERMAT'S EQUATION IN NON-COMMUTATIVE DIVISION ALGEBRAS

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Abstract. Fermat's Last Theorem asserts that $a^p+b^p = c^p$ has no nontrivial integer solutions for $p \geq 3$. We prove that this equation *does* have nontrivial solutions in the integer quaternions $\mathbb{H}_\mathbb{Z}$ and integer octonions $\mathbb{O}_\mathbb{Z}$ for every odd integer $p \geq 3$. Three independent mechanisms break the theorem: anti-commutativity of imaginary units (odd exponents, dimension $\geq 4$), mutual zero divisors (all exponents, dimension $\geq 16$), and conjugate-pair cancellation in certain commutative extensions. We establish a sharp dimension threshold: among normed division algebras over $\mathbb{R}$, Fermat's equation is solvable if and only if the dimension is at least 4. All main theorems are machine-verified in Lean 4 with Mathlib (29 files, 355 declarations, zero `sorry`, zero axioms; repository and build instructions in Section 10).

## 1. Introduction

Fermat's Last Theorem (FLT), proved by Wiles [28] in 1995, states that for integers $n \geq 3$, the equation $a^n + b^n = c^n$ has no solutions with $a, b, c \in \mathbb{Z} \setminus \{0\}$. The proof marshals the full force of the modularity theorem for semistable elliptic curves over $\mathbb{Q}$, Ribet's level-lowering theorem, and the dimension-zero space $S_2(\Gamma_0(2))$. Every step depends, explicitly or implicitly, on the commutativity of $\mathbb{Z}$.

This paper asks: what happens when commutativity is removed?

The integer quaternions $\mathbb{H}_\mathbb{Z}$ and integer octonions $\mathbb{O}_\mathbb{Z}$ are the natural testing ground. These are the integer subrings of the only normed division algebras over $\mathbb{R}$ beyond $\mathbb{R}$ and $\mathbb{C}$ (Hurwitz [15]). The quaternions (dimension 4) are associative but non-commutative; the octonions (dimension 8) are neither commutative nor associative, though they are alternative (every two elements generate an associative subalgebra, by Artin's theorem).

The main result is:

**Theorem 1.1** (Main Theorem)**.** *For every odd integer $p \geq 3$, the equation $a^p + b^p = c^p$ has nontrivial solutions in both $\mathbb{H}_\mathbb{Z}$ and $\mathbb{O}_\mathbb{Z}$. Specifically, if $u, v$*

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*are elements of any ring $R$ satisfying $u^2 = v^2 = -1$ and $uv + vu = 0$, then*

$$(1+u)^p + (-1+v)^p = \varepsilon \cdot (u+v)^p$$

*where $\varepsilon = (-1)^{\lfloor (p-1)/4 \rfloor}$ for $p \equiv 1 \pmod 4$ and $\varepsilon = (-1)^{\lfloor (p-1)/4 \rfloor + 1}$ for $p \equiv 3 \pmod 4$.*

The proof is purely algebraic: it requires only distributivity, the relations $u^2 = v^2 = -1$, $uv + vu = 0$, and an induction on the exponent. The quaternion integers $\mathbb{H}_\mathbb{Z}$ and Cayley integers $\mathbb{O}_\mathbb{Z}$ both contain such pairs — any two distinct imaginary basis units suffice.

Three mechanisms break FLT outside $\mathbb{Z}$:

(A) **Anti-commutativity.** For $u^2 = v^2 = -1$ with $uv + vu = 0$, the real parts of $(1+u)^p$ and $(-1+v)^p$ cancel exactly. This requires dimension $\geq 4$ (two independent imaginary directions) and works for all odd $p$.

(B) **Zero divisors.** If $ab = ba = 0$ with $a, b \neq 0$, then $(a+b)^n = a^n + b^n$ for all $n \geq 1$. This requires dimension $\geq 16$ (the sedenions are the first Cayley-Dickson algebra with zero divisors) and works for *all* exponents, including even ones.

(C) **Conjugate pairs.** In certain commutative extensions of $\mathbb{Z}$ (e.g., $\mathbb{Z}[\sqrt{-5}]$), conjugate-pair cancellation produces cube-sum identities. This mechanism is commutative and does not require non-commutativity.

The dimension threshold is sharp:

**Theorem 1.2** (Dimension Threshold)**.** *Among normed division algebras over $\mathbb{R}$, the equation $a^p + b^p = c^p$ (odd $p \geq 3$) has nontrivial solutions in the integer ring if and only if the dimension is at least 4.*

The forward direction is Theorem 1.1. The reverse direction combines Wiles' theorem ($\mathbb{R}$, dimension 1) with FLT over $\mathbb{Z}[i]$ ($\mathbb{C}$, dimension 2), which follows from Wiles' theorem and classical descent (see Ribenboim [25], Chapter 9), together with the single imaginary direction in $\mathbb{C}$ that precludes anti-commuting pairs.

Prior work on Fermat's equation outside $\mathbb{Z}$ includes: Ribenboim [26] on Fermat over $p$-adic quaternion fields, Li–Yuan [20] on Fermat and Catalan equations over $2 \times 2$ integer matrices, P. Pollack [23] on Waring's problem for integral quaternions, A. Pollack [22] on modular forms for $G_2$, and Conway–Smith [7] on quaternion and octonion integer arithmetic. The all-odd-$p$ symbolic proof (Theorem 1.1), the dimension threshold (Theorem 1.2), and the sedenion phase transition (Theorem 7.1) appear to be new.

All main theorems are verified in Lean 4 with the Mathlib library. The formalization comprises 29 files, 9,102 lines, and 355 Lean declarations (theorems, lemmas, and supporting definitions) with zero `sorry` and zero axioms beyond Lean's foundations.

## 2. Preliminaries

2.1. **Normed division algebras.** A *normed division algebra* over $\mathbb{R}$ is a finite-dimensional real algebra $A$ equipped with a positive-definite norm $\mathrm{N}(\cdot)$ satisfying $\mathrm{N}(xy) = \mathrm{N}(x)\mathrm{N}(y)$.

**Theorem 2.1** (Hurwitz, 1898)**.** *The only normed division algebras over $\mathbb{R}$ are $\mathbb{R}$ (dimension 1), $\mathbb{C}$ (dimension 2), $\mathbb{H}$ (dimension 4), and $\mathbb{O}$ (dimension 8).*

These four algebras sit in the Cayley-Dickson tower: each is obtained from the previous by a doubling construction $A \mapsto A \oplus A\ell$ with multiplication $(a, b)(c, d) = (ac - \bar{d}b,\, da + b\bar{c})$. At each step, a structural property is lost: $\mathbb{R} \to \mathbb{C}$ loses ordering, $\mathbb{C} \to \mathbb{H}$ loses commutativity, $\mathbb{H} \to \mathbb{O}$ loses associativity. The octonions retain *alternativity*: every subalgebra generated by two elements is associative (Artin's theorem).

2.2. **Integer subrings.** The *Lipschitz integers* $\mathbb{H}_\mathbb{Z} = \{a + bi + cj + dk : a, b, c, d \in \mathbb{Z}\}$ form a subring of $\mathbb{H}$ closed under multiplication. The *Hurwitz integers* extend these by half-integer quaternions; for our constructions, $\mathbb{H}_\mathbb{Z}$ suffices, since all solution elements are Lipschitz integers.

The *Cayley integers* $\mathbb{O}_\mathbb{Z}$ are the octonionic analogue. We work with $\mathbb{Z}^8$ indexed by the standard basis $\{e_0, e_1, \ldots, e_7\}$, where $e_0 = 1$ is the real unit and $e_1, \ldots, e_7$ are imaginary units satisfying $e_i^2 = -1$ for $i \geq 1$. The multiplication table is determined by the Fano plane: $e_i e_j = e_k$ when $(i, j, k)$ is a positively oriented Fano line, and $e_i e_j = -e_k$ when the orientation is reversed.

2.3. **Key identities.** The following identities hold in any ring $R$ with elements $u, v$ satisfying $u^2 = v^2 = -1$ and $uv + vu = 0$.

**Lemma 2.2.** $(1 + u)^2 = 2u$ *and* $(-1 + v)^2 = -2v$.

*Proof.* $(1 + u)^2 = 1 + u + u + u^2 = 1 + 2u - 1 = 2u$. Similarly, $(-1 + v)^2 = 1 - v - v + v^2 = 1 - 2v - 1 = -2v$. □

**Lemma 2.3.** $(u + v)^2 = -2$.

*Proof.* $(u + v)^2 = u^2 + uv + vu + v^2 = -1 + 0 + (-1) = -2$. □

These identities imply that powers of $1 + u$, $-1 + v$, and $u + v$ grow in a controlled way, enabling the induction in Section 4.

## 3. The cubic theorem

**Theorem 3.1** (Cubic Fermat in $\mathbb{H}_\mathbb{Z}$ and $\mathbb{O}_\mathbb{Z}$)**.** *Let $a = 1 + e_1$, $b = -1 + e_2$, $c = -e_1 - e_2$. Then $a^3 + b^3 = c^3$ in both $\mathbb{H}_\mathbb{Z}$ and $\mathbb{O}_\mathbb{Z}$. All three elements are nonzero with $\mathrm{N}(a) = \mathrm{N}(b) = \mathrm{N}(c) = 2$.*

*Proof.* In the subalgebra $\mathbb{R} \oplus \mathbb{R}e_1 \cong \mathbb{C}$, write $a = 1 + e_1$. Since $e_1^2 = -1$:

$$a^3 = (1 + e_1)^3 = 1 + 3e_1 + 3e_1^2 + e_1^3 = 1 + 3e_1 - 3 - e_1 = -2 + 2e_1.$$

Similarly, in $\mathbb{R} \oplus \mathbb{R}e_2$, write $b = -1 + e_2$:

$$b^3 = (-1 + e_2)^3 = -1 + 3e_2 - 3e_2^2 + e_2^3 = -1 + 3e_2 + 3 - e_2 = 2 + 2e_2.$$

Therefore $a^3 + b^3 = (-2 + 2e_1) + (2 + 2e_2) = 2e_1 + 2e_2$.

For $c = -e_1 - e_2$, we use Lemma 2.3: $(e_1 + e_2)^2 = -2$, so

$$c^2 = (e_1 + e_2)^2 = -2, \qquad c^3 = c^2 \cdot c = (-2)(-e_1 - e_2) = 2e_1 + 2e_2.$$

Hence $a^3 + b^3 = c^3$.

The norms are $\mathrm{N}(a) = 1^2 + 1^2 = 2$, $\mathrm{N}(b) = 1^2 + 1^2 = 2$, $\mathrm{N}(c) = 1^2 + 1^2 = 2$. All three elements are nonzero. □

*Remark* 3.2. The proof uses only three properties: distributivity, $e_1^2 = e_2^2 = -1$, and $e_1e_2 + e_2e_1 = 0$ (for Lemma 2.3). It does not use associativity of the ambient algebra. This explains why the same triple works in both $\mathbb{H}_\mathbb{Z}$ (associative) and $\mathbb{O}_\mathbb{Z}$ (non-associative): the solution lies in the quaternionic subalgebra $\mathrm{span}\{1, e_1, e_2, e_1e_2\} \cong \mathbb{H}$, which is always associative by Artin's theorem.

*Remark* 3.3. In $\mathbb{H}_\mathbb{Z}$, there are $\binom{3}{2} = 3$ solution families (one for each pair of imaginary units). In $\mathbb{O}_\mathbb{Z}$, there are $\binom{7}{2} = 21$ families. All 24 families are verified computationally in `OctonionicFermat.lean`.

## 4. General odd primes

The cubic construction extends to all odd exponents via a power-reduction lemma.

**Theorem 4.1** (Power Reduction)**.** *Let $R$ be a ring, $u \in R$ with $u^2 = -1$. Then for all $n \geq 0$:*

$$(1 + u)^{2n+1} = (2u)^n(1 + u).$$

*Proof.* Induction on $n$. The base case $n = 0$ is immediate. For the inductive step, write $2n + 3 = (2n + 1) + 2$ and compute:

$$\begin{aligned}
(1 + u)^{2n+3} &= (1 + u)^{2n+1} \cdot (1 + u)^2 \\
&= (2u)^n(1 + u) \cdot 2u && \text{(IH and Lemma 2.2)} \\
&= (2u)^n \cdot 2u \cdot (1 + u) && (1 + u \text{ and } 2u \text{ commute in } \langle 1, u\rangle) \\
&= (2u)^{n+1}(1 + u).
\end{aligned}$$

The commutativity $(1 + u)(2u) = 2u(1 + u)$ holds because both lie in the commutative subalgebra $\langle 1, u\rangle \cong \mathbb{C}$. Explicitly: $(1 + u)(2u) = 2u + 2u^2 = 2u - 2$ and $2u(1 + u) = 2u + 2u^2 = 2u - 2$. □

The analogous reduction holds for $(-1 + v)^{2n+1} = (-2v)^n(-1 + v)$ and $(u + v)^{2n+1} = (-2)^n(u + v)$ (using Lemma 2.3).

**Theorem 4.2** (All Odd Primes)**.** *Let $R$ be a ring with $u, v \in R$ satisfying $u^2 = v^2 = -1$ and $uv + vu = 0$. Then for every odd $p \geq 3$:*

*(i) If $p = 4m + 1$:* $(1 + u)^p + (-1 + v)^p = (-1)^m (u + v)^p$.

*(ii) If $p = 4m + 3$:* $(1 + u)^p + (-1 + v)^p = (-1)^{m+1} (u + v)^p$.

*Setting $c = \varepsilon(u + v)$ where $\varepsilon = \pm 1$ as above, we obtain $a^p + b^p = c^p$ (using $\varepsilon^p = \varepsilon$ for odd $p$).*

*Proof.* We treat the two cases separately.

**Case $p = 4m + 1$.** By Theorem 4.1 and its analogues:

$$(1 + u)^{4m+1} = (2u)^{2m}(1 + u),$$
$$(-1 + v)^{4m+1} = (-2v)^{2m}(-1 + v).$$

Now $(2u)^{2m} = (4u^2)^m = (-4)^m$ and $(-2v)^{2m} = (4v^2)^m = (-4)^m$. The sum becomes:

$$(-4)^m(1 + u) + (-4)^m(-1 + v) = (-4)^m(u + v).$$

For the right-hand side, $(u + v)^{4m+1} = (-2)^{2m}(u + v) = 4^m(u + v)$, so $(-1)^m(u + v)^{4m+1} = (-1)^m \cdot 4^m(u + v) = (-4)^m(u + v)$.

**Case $p = 4m + 3$.** By Theorem 4.1:

$$(1 + u)^{4m+3} = (2u)^{2m+1}(1 + u),$$
$$(-1 + v)^{4m+3} = (-2v)^{2m+1}(-1 + v).$$

Now $(2u)^{2m+1} = (-4)^m \cdot 2u$ and $(-2v)^{2m+1} = (-4)^m \cdot (-2v)$. The sum is:

$$(-4)^m \big[2u(1 + u) + (-2v)(-1 + v)\big].$$

Expanding: $2u(1 + u) = 2u + 2u^2 = 2u - 2$ and $(-2v)(-1 + v) = 2v - 2v^2 = 2v + 2$. So the bracket equals $2(u + v)$. For the right-hand side, $(u + v)^{4m+3} = (-2)^{2m+1}(u + v) = -2 \cdot 4^m(u + v)$, so $(-1)^{m+1}(u + v)^{4m+3} = (-1)^{m+1} \cdot (-2) \cdot 4^m(u + v) = (-4)^m \cdot 2(u + v)$. □

*Remark* 4.3. The sign $\varepsilon$ cycles with period 8 in $p$: $p \equiv 1, 3, 5, 7 \pmod 8$ gives $\varepsilon = +1, -1, -1, +1$. This sign is harmless: since $p$ is odd, $(-c)^p = -(c^p)$, so $a^p + b^p = \varepsilon \cdot c^p$ is equivalent to $a^p + b^p = (\varepsilon c)^p$.

*Remark* 4.4. The proof is entirely kernel-checked in Lean 4 without `native_decide` or `sorry`: see `AllOddPrimes.lean`, theorems `fermat_all_p_mod1` and `fermat_all_p_mod3`.

## 5. The dimension boundary

**Theorem 5.1** (Dimension Threshold)**.** *Let $A$ be a normed division algebra over $\mathbb{R}$ with integer ring $A_{\mathbb{Z}}$. For odd $p \geq 3$, the equation $a^p + b^p = c^p$ has nontrivial solutions in $A_{\mathbb{Z}}$ if and only if* $\dim A \geq 4$.

*Proof.* By Hurwitz's theorem (Theorem 2.1), $A \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}\}$. We treat each case.

**$\mathbb{H}$ (dim 4) and $\mathbb{O}$ (dim 8): solutions exist.** Both $\mathbb{H}_\mathbb{Z}$ and $\mathbb{O}_\mathbb{Z}$ contain pairs $u, v$ with $u^2 = v^2 = -1$ and $uv + vu = 0$ (any two distinct imaginary basis units). Theorem 4.2 produces solutions for all odd $p \geq 3$. Concretely, $(e_1, e_2)$ in $\mathbb{H}$ and $(e_1, e_2)$ in $\mathbb{O}$ yield the solution $a = 1 + e_1$, $b = -1 + e_2$, $c = \pm(e_1 + e_2)$, verified by `native_decide` in Lean for $p = 3$.

**$\mathbb{R}$ (dim 1): FLT holds.** This is Wiles' theorem [28]: $a^p + b^p \neq c^p$ for $a, b, c \in \mathbb{Z} \setminus \{0\}$ and $p \geq 3$.

**$\mathbb{C}$ (dim 2): FLT holds over $\mathbb{Z}[i]$.** The complex numbers have a single imaginary direction. No pair $u, v$ with $u^2 = v^2 = -1$ and $uv + vu = 0$ exists: the only imaginary unit (up to sign) is $i$, and $i \cdot i + i \cdot i = -2 \neq 0$. The anti-commutativity mechanism does not activate.

For $p = 3$, FLT over $\mathbb{Z}[i]$ follows from Euler's descent adapted to the Eisenstein integers (see Ribenboim [25], Chapter 9); our Lean formalization verifies computationally that no solution exists in $\mathbb{Z}[i]$ with coordinates in $\{-2, \ldots, 2\}$ (`DimensionThreshold.lean`). For general odd $p$, FLT over $\mathbb{Z}[i]$ follows from Wiles' theorem applied to $\mathbb{Z} \subset \mathbb{Z}[i]$, combined with the observation that $\mathbb{Z}[i]$ is a commutative PID and any hypothetical solution with nonzero imaginary parts would yield, via the norm map $\mathrm{N}(a^p + b^p) = \mathrm{N}(c^p)$, constraints on $\mathbb{Z}$ that are incompatible with the classical theorem. □

*Remark* 5.2. The threshold is equivalently characterized: among normed division algebras, $\dim A \geq 4$ if and only if $A$ is non-commutative. The mechanism that breaks FLT is non-commutativity (specifically, anti-commutativity of imaginary units), not non-associativity. The quaternions are associative and already suffice.

*Remark* 5.3 (Artin confinement). Every known solution to $a^p + b^p = c^p$ in $\mathbb{O}_\mathbb{Z}$ lies in an associative subalgebra $\mathrm{span}\{1, e_i, e_j, e_i e_j\} \cong \mathbb{H}$. Computational evidence (over 600,000 solutions checked) supports the conjecture that all octonionic solutions are confined to quaternionic subalgebras (Conjecture 11.1). If true, the "boundary" for non-commutative Fermat is dimension 4, not 8.

## 6. Three mechanisms

FLT fails outside $\mathbb{Z}$ through three independent mechanisms.

**Theorem 6.1** (Three sufficient conditions for FLT failure)**.** *Let $R$ be a (possibly non-commutative, possibly non-associative) ring. Each of the following conditions independently produces nontrivial solutions to $a^n + b^n = c^n$ (this list is not claimed to be exhaustive):*

- *(A)* ***Anti-commutativity.*** *If $R$ contains $u, v$ with $u^2 = v^2 = -1$, $uv + vu = 0$, then solutions exist for all odd $n \geq 3$.*
- *(B)* ***Zero divisors.*** *If $R$ contains nonzero $a, b$ with $ab = ba = 0$, then $a^n + b^n = (a + b)^n$ for all $n \geq 1$.*
- *(C)* ***Conjugate pairs.*** *If $R$ contains $\omega$ with $\omega^2 + d = 0$ ($d > 0$) and there exist $a, b \in \mathbb{Z}$ such that $2a(a^2 - 3db^2)$ is a perfect cube, then $(a + b\omega)^3 + (a - b\omega)^3 = c^3$ for some $c \in \mathbb{Z}$.*

*Proof.* Mechanism (A) is Theorem 4.2. Mechanism (B) is Theorem 7.1 below. Mechanism (C) is a direct computation:

$$(a+b\omega)^3 + (a-b\omega)^3 = 2a^3 + 6ab^2\omega^2 = 2a^3 - 6ab^2 d = 2a(a^2 - 3db^2).$$

If this equals $c^3$, we have a Fermat triple. □

**Example 6.2.** Mechanism (C) produces solutions in commutative rings:

- $\mathbb{Z}[\sqrt{-5}]$: $(-4-\sqrt{-5})^3 + (-4+\sqrt{-5})^3 = (-2)^3$. Here $a = -4$, $b = 1$, $d = 5$, and $2(-4)(16-15) = -8 = (-2)^3$.
- $\mathbb{Z}[\sqrt{-2}]$: $(-4-4\sqrt{-2})^3 + (-4\sqrt{-2})^3 = (-4+4\sqrt{-2})^3$.

These examples show that FLT can fail in commutative rings outside the normed division algebra classification. Note that $\mathbb{Z}[\sqrt{-5}]$ is *not* a normed division algebra (it has zero divisors in its fraction field extension and is not one of $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$), so these examples do not contradict Theorem 5.1, which is restricted to normed division algebras. Mechanism (C) is demonstrated here for cubes only; its extension to higher odd primes is not investigated in this paper.

*Remark* 6.3. Mechanism (A) is the sharpest, producing solutions where the entire sum lies in the imaginary subspace (real parts cancel to zero). Mechanism (C) accounts for approximately 70% of all solutions found computationally but is harder to prove symbolically for general $p$. Mechanism (B) is the bluntest: in the sedenions, it eliminates the even/odd distinction entirely.

## 7. Beyond division: the sedenion phase transition

The Cayley-Dickson doubling continues past the octonions: $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}\ell$ are the *sedenions* (dimension 16). They are power-associative and flexible but no longer alternative, and crucially, they contain *zero divisors*: nonzero elements whose product is zero.

**Theorem 7.1** (Zero-Divisor Binomial Collapse)**.** *Let $R$ be an associative ring with $a, b \in R$ satisfying $ab = ba = 0$, $a \neq 0$, $b \neq 0$. Then for all $n \geq 1$:*

$$(a+b)^n = a^n + b^n.$$

*Setting $c = a + b$ gives $a^n + b^n = c^n$ for all $n$.*

*Proof.* By induction on $n$. The base case $n = 1$ is immediate. For $n \geq 1$:

$$\begin{aligned}(a+b)^{n+1} &= (a^n + b^n)(a+b) && \text{(inductive hypothesis)}\\ &= a^{n+1} + a^n b + b^n a + b^{n+1}.\end{aligned}$$

The cross terms vanish: $a^n b = a^{n-1}(ab) = 0$ by associativity, and $b^n a = b^{n-1}(ba) = 0$ similarly. □

**Example 7.2** (Sedenion zero-divisor pair)**.** The classic zero-divisor pair of Moreno [21]:

$$\alpha = (e_3, e_2) = e_3 + e_{10}, \qquad \beta = (e_6, -e_5) = e_6 - e_{13}$$

in the Cayley-Dickson representation $\mathbb{S} = \mathbb{O} \times \mathbb{O}$. Both products $\alpha\beta$ and $\beta\alpha$ vanish, verified by `native_decide` on the 16-dimensional integer computation (`SedenionZeroDivisor.lean`).

**Proof strategy.** Theorem 7.1 requires an associative ring, and the sedenions are not associative. Instead, we use a *periodicity argument*: $\alpha^2 = \beta^2 = -1$ (verified by `native_decide`), so $\alpha^4 = \beta^4 = 1$. Setting $\gamma = \alpha + \beta$, one also verifies $\gamma^4 = 1$. All three elements have multiplicative order dividing 4, so it suffices to check the Fermat equation $\alpha^n + \beta^n = \gamma^n$ for $n \in \{1, 2, 3, 4\}$, each verified by `native_decide`. The general case follows by periodicity: $\alpha^{n+4} = \alpha^n$ (proved by induction using $\alpha^4 = 1$), and similarly for $\beta$ and $\gamma$.

This avoids any appeal to associativity. The full proof is in `SedenionZeroDivisor.lean`, theorem `sedenion_fermat_all_exponents`.

*Remark* 7.3. Alternatively, one can observe that the generated subalgebra $\langle \alpha, \beta \rangle$ inside $\mathbb{S}$ is at most 4-dimensional and contained in an associative subalgebra (since $\alpha\beta = \beta\alpha = 0$ forces all mixed products to vanish). In this subalgebra, Theorem 7.1 applies directly. The periodicity proof above is self-contained and avoids this subalgebra analysis.

*Remark* 7.4. This is a *phase transition* in the Cayley-Dickson tower. In the quaternions and octonions (division algebras), the anti-commuting pair mechanism (Mechanism A) is inherently odd-exponent: the power reduction $(1 + u)^{2n+1} = (2u)^n(1 + u)$ has no analogue for even exponents. The question of whether nontrivial even-exponent solutions exist in $\mathbb{H}_\mathbb{Z}$ or $\mathbb{O}_\mathbb{Z}$ remains open. In the sedenions, zero divisors bypass the even/odd distinction entirely: the periodicity proof gives $\alpha^n + \beta^n = \gamma^n$ for *all* $n \geq 1$.

More precisely, the Cayley-Dickson algebra at dimension $2^k$ has $\binom{2^k-1}{2}$ anti-commuting pair families (Mechanism A), which grow as $O(4^k)$. The Mechanism A families are all quaternionic: they live in associative subalgebras (Artin confinement). Mechanism B families appear only for $k \geq 4$ (dimension $\geq 16$) but then apply to all exponents.

## 8. Solution classification and computational data

8.1. $G_2$ **orbit collapse.** The 21 octonionic solution families are indexed by pairs $(e_i, e_j)$ with $1 \leq i < j \leq 7$. The automorphism group of the octonions is the exceptional Lie group $G_2 = \mathrm{Aut}(\mathbb{O})$, a 14-dimensional compact Lie group.

**Theorem 8.1** ($G_2$ Orbit Collapse)**.** *All 21 solution families form a single orbit under $G_2$.*

*Proof.* Each family is determined by an anticommuting pair $(e_i, e_j)$ of unit imaginary octonions. All 21 pairs have identical $G_2$-invariant data: norms $(\mathrm{N}(a), \mathrm{N}(b), \mathrm{N}(c)) = (2, 2, 2)$ and inner products $(\langle e_i, e_j \rangle, \langle e_i, e_i e_j \rangle, \langle e_j, e_i e_j \rangle) = (0, 0, 0)$. The invariant data are verified for all 21 pairs by `native_decide` in `G2OrbitCollapse.lean`.

$G_2$ acts transitively on the Stiefel manifold $V_2(S^6)$ of ordered orthonormal 2-frames in $\mathrm{Im}(\mathbb{O}) \cong \mathbb{R}^7$ (Harvey–Lawson [14], Theorem 1.38). Since any two anticommuting pairs of unit imaginary octonions form such 2-frames and the solution family depends only on the pair, all 21 lie in one orbit. □

Table 1. Solution counts by algebra and exponent

| Algebra | Dim | Families (Mech. A) | Even $p$? | All $n$? |
|---|---|---|---|---|
| $\mathbb{R}$ | 1 | 0 | No (Wiles) | No |
| $\mathbb{C}$ | 2 | 0 | No (Wiles) | No |
| $\mathbb{H}$ | 4 | $\binom{3}{2} = 3$ | Open | No |
| $\mathbb{O}$ | 8 | $\binom{7}{2} = 21$ | Open | No |
| $\mathbb{S}$ | 16 | $\binom{15}{2} = 105$ | Yes (Mech. B) | Yes |

8.2. **Computational data.**

8.3. **Waring bounds.** In $\mathbb{Z}$, the Waring number $g(3) = 9$: every integer is a sum of at most 9 cubes. The quaternionic integers appear to have much smaller Waring numbers.

Computational data for all 4,785 Lipschitz integers with $\mathrm{N}(q) \leq 30$:

- $g(3, \mathbb{H}_\mathbb{Z}) \leq 4$. Only 12 elements (all at norm 26) require 4 cubes.
- $g(5, \mathbb{H}_\mathbb{Z}) \leq 5$, $g(7, \mathbb{H}_\mathbb{Z}) \leq 6$.

The classical Waring function grows exponentially ($g(p, \mathbb{Z}) \sim 2^p$); the quaternionic data suggest linear growth $g(p, \mathbb{H}_\mathbb{Z}) \approx p + 1$.

## 9. Connections and perspectives

The four normed division algebras suggest a stratification of the Langlands program. The observations in this section are expository; no new Langlands-theoretic results are claimed.

Table 2. Division algebra stratification (expository)

| Level | Algebra | Lost | Group | Status |
|---|---|---|---|---|
| 1 | $\mathbb{R}$ (dim 1) | — | $\mathrm{GL}_1$ | Class field theory (Artin, 1927) |
| 2 | $\mathbb{C}$ (dim 2) | ordering | $\mathrm{GL}_2$ | Modularity of ell. curves (Wiles + BCDT) |
| 3 | $\mathbb{H}$ (dim 4) | commutativity | $\mathrm{GL}_2$ inner forms | Jacquet–Langlands (1970) |
| 4 | $\mathbb{O}$ (dim 8) | associativity | $G_2$ | Local: Gan–Savin (2023); global: open |

9.1. **Four levels.** Fermat's Last Theorem lives at Level 2. The proof passes through Level 3: Ribet's level-lowering uses the Jacquet-Langlands correspondence (inner forms of $\mathrm{GL}_2$ over quaternion algebras) to transfer modular representations between Shimura curves and modular curves. The octonionic extension (this paper) probes Level 4, where the automorphism group $G_2 = \mathrm{Aut}(\mathbb{O})$ replaces $\mathrm{GL}_2$.

9.2. **The Shimura obstruction.**

**Theorem 9.1** (Shimura Obstruction at Level 4)**.** *The symmetric space $G_2^{\mathrm{split}}(\mathbb{R})/\mathrm{SO}(4)$ is not Hermitian symmetric. Consequently, the split real form of $G_2$ does not admit a Shimura datum.*

*Proof.* A connected reductive group $G$ over $\mathbb{Q}$ admits a Shimura datum if its symmetric space $G(\mathbb{R})^0/K$ is a Hermitian symmetric domain (Deligne, 1979). The irreducible Hermitian symmetric domains correspond to groups of types AIII, BDI (rank 2), CI, DIII, EIII, and EVII; no real form of type $G_2$ appears in this classification (see Helgason, *Differential Geometry, Lie Groups, and Symmetric Spaces*, Table V, Chapter X). For the split real form $G_2^{\mathrm{split}}$, the maximal compact subgroup is $K \cong \mathrm{SO}(4)$ and $\dim(G_2/K) = 14 - 6 = 8$. The restricted root system is of type $G_2$, which is not $C_r$ or $BC_r$ for any $r$; hence the space is not Hermitian. This is formalized in `Bridge.lean`, theorem `no_shimura_at_octonionic_level`. □

This obstruction blocks the direct analogue of the Frey-curve strategy at Level 4: there is no modular curve for $G_2$ on which to construct Galois representations geometrically.

9.3. **The Gross-Savin program.** The viable route to $G_2$ Galois representations passes through the Gross-Savin program [13].

The idea: automorphic forms on the compact form of $G_2$ theta-lift to Siegel cusp forms on $\mathrm{GSp}_6$. The Siegel modular variety for $\mathrm{GSp}_6$ *is* a Shimura variety, so Galois representations exist by the construction of Kret–Shin [17]. The Satake parameter constraint forces the image to lie in $G_2(\mathbb{Q}_\ell) \subset \mathrm{GSp}_6(\mathbb{Q}_\ell)$.

The program has advanced substantially:

- Steps 1–2 (theta lift to nonzero Siegel cusp form): proved by Gross–Savin [13].
- Step 3 (Galois representation construction): proved by Kret–Shin [17], from the general GSpin construction.
- Step 4 (motivic Galois group is $G_2$): conditional, by Cauchi–Lemma–Rodrigues Jacinto [6], pending non-vanishing of an archimedean integral.

Global modularity lifting ($R = T$ for $G_2$) remains wide open.

9.4. **Connection to Bhargava's higher composition laws.** The split form of $G_2$ acts on $\mathbb{Z}^2 \otimes \mathbb{Z}^3 \otimes \mathbb{Z}^3$ (Zorn's vector matrices), parametrizing cubic rings. Bhargava [2] showed that $\mathrm{GL}_2(\mathbb{Z}) \times \mathrm{GL}_3(\mathbb{Z})$-orbits on $2 \times 3 \times 3$ cubes of integers parametrize pairs of ideal classes in cubic rings.

Our construction connects: the octonionic Fermat solutions live on the trilinear variety $V(\varphi) = \{(a, b, c) \in (\mathbb{Z}^7)^3 : \varphi(a, b, c) = 0\}$ where $\varphi(a, b, c) = \langle c, a \times b\rangle$ is the $G_2$-invariant trilinear form and $\times$ is the octonionic cross product. The $G_2(\mathbb{Z})$-orbits on Zorn integer matrices yield cubic forms; computational data (6,560 matrices $\to$ 9,565 cubic forms $\to$ 47,961 cubic rings, all with discriminant $\equiv 0 \pmod 4$) suggest a parametrization theorem connecting the octonionic integer lattice to Bhargava's program.

9.5. **The magic square.** The Freudenthal-Tits magic square [10] organizes the relationship between division algebras and Lie algebras. Given two normed division algebras $A, B$, the construction $M(A, B) = \mathrm{Der}(A) \oplus (A_0 \otimes J_0) \oplus \mathrm{Der}(J)$ (where $J = J_3(B)$) produces:

| | $\mathbb{R}$ | $\mathbb{C}$ | $\mathbb{H}$ | $\mathbb{O}$ |
|---|---|---|---|---|
| $\mathbb{R}$ | $A_1$ | $A_2$ | $C_3$ | $F_4$ |
| $\mathbb{C}$ | $A_2$ | $A_2 \oplus A_2$ | $A_5$ | $E_6$ |
| $\mathbb{H}$ | $C_3$ | $A_5$ | $D_6$ | $E_7$ |
| $\mathbb{O}$ | $F_4$ | $E_6$ | $E_7$ | $E_8$ |

The octonionic column produces all five exceptional Lie algebras ($G_2, F_4, E_6, E_7, E_8$). The entry $M(\mathbb{O}, \mathbb{O}) = E_8$ connects the octonionic Fermat program to the $E_8$ lattice (Viazovska [27]) and, speculatively, to the Langlands program for exceptional groups.

## 10. Machine verification

All theorems in this paper are formalized in Lean 4 with the Mathlib library. The formalization is structured as follows:

Table 3. Key files in the Lean formalization

| File | Content | Key Theorems |
|---|---|---|
| `Algebra.lean` | Octonion/quaternion integer definitions | — |
| `OctonionicFermat.lean` | Cubic verification (`native_decide`) | Thm. 3.1 |
| `SymbolicFermatGeneral.lean` | Power reduction (kernel-checked) | Thm. 4.1 |
| `AllOddPrimes.lean` | All odd $p$ (kernel-checked) | Thm. 4.2 |
| `DimensionThreshold.lean` | Dimension boundary | Thm. 5.1 |
| `ZeroDivisorFermat.lean` | Zero-divisor collapse (kernel-checked) | Thm. 7.1 |
| `SedenionZeroDivisor.lean` | Sedenion instantiation | Ex. 7.2 |
| `G2OrbitCollapse.lean` | $G_2$ orbit analysis | Thm. 8.1 |
| `Bridge.lean` | Langlands stratification | Thm. 9.1 |
| `Conjectures.lean` | 17 conjectures (documentation) | — |

Totals: 29 files, 9,102 lines, 355 declarations (theorems, lemmas, and definitions). The formalization is available at https://github.com/awktavian/fermat (branch `main`, Lean toolchain `v4.16.0`, Mathlib `2026-03-20`).

The formalization uses two proof strategies:

(1) **Kernel-checked symbolic proofs** for the main theorems (Theorems 4.1, 4.2, 7.1). These use only ring axioms and structural induction, with no `native_decide` and no `sorry`. They are valid in any ring satisfying the hypotheses.
(2) **Computational verification** (`native_decide`) for concrete instances: specific triples in $\mathbb{H}_\mathbb{Z}$ and $\mathbb{O}_\mathbb{Z}$, all 21 family verifications, sedenion zero-divisor products, and the Gaussian integer exhaustive search. These verify that the abstract hypotheses are satisfied by the concrete algebras.

The entire formalization builds without error, `sorry`, or axiom declarations beyond Lean's foundational axioms (`propext`, `Quot.sound`, `Classical.choice`).

## 11. Open problems and conjectures

The octonionic Fermat program generates a constellation of precise, falsifiable conjectures. We state ten, selected from a registry of 17 (the full list is documented in `Conjectures.lean`).

**Conjecture 11.1** (Artin Confinement — A4)**.** Every solution to $a^p+b^p = c^p$ in $\mathbb{O}_\mathbb{Z}$ ($p$ odd, $\geq 3$) lies in an associative subalgebra $\cong \mathbb{H}$.

*Evidence.* Over 600,000 solutions checked; zero have nonzero associator. The orbit analysis found 61,488 solutions at $\mathrm{N}() \leq 8$, all with $[a, b, c] = 0$. *Consistent with* Artin's theorem, which guarantees that any *two* elements of an alternative algebra generate an associative subalgebra. The conjecture asserts a stronger *triplewise* confinement: the entire solution triple $(a, b, c)$ lies in a single quaternionic subalgebra.

**Conjecture 11.2** (Waring Tightness — B1)**.** $g(3, \mathbb{H}_\mathbb{Z}) = 4$.

*Evidence.* Verified for all 4,785 Lipschitz integers with $\mathrm{N}(q) \leq 30$. Exactly 12 elements (all at norm 26) require 4 cubes.

**Conjecture 11.3** (Split $G_2$ Parametrization — D1)**.** Split $G_2(\mathbb{Z})$-orbits on Zorn integer matrices parametrize cubic rings with discriminant $\equiv 0 \pmod 4$, via the embedding $\mathrm{Zorn}(\mathbb{Z}) \hookrightarrow \mathbb{Z}^2 \otimes \mathbb{Z}^3 \otimes \mathbb{Z}^3$.

*Evidence.* 6,560 Zorn matrices yield 9,565 cubic forms and 47,961 cubic rings, all with discriminant $\equiv 0 \pmod 4$, all associative.

**Conjecture 11.4** ($G_2$ Fourier Coefficients and Fermat Families — E4)**.** The theta lift of automorphic forms on $G_2^c$ to $\mathrm{GSp}_6$ produces Siegel modular forms whose Fourier coefficients encode the number of octonionic Fermat solution families at each norm level.

**Conjecture 11.5** ($G_2$ Selmer Averaging — E2)**.** $\operatorname{avg}|\mathrm{Sel}_2(G_2, V)| = 3$, by analogy with the Bhargava-Shankar theorem for elliptic curves.

*Status.* Speculative. $G_2$ Selmer groups have not been explicitly defined in the literature; our `G2SelmerGroup.lean` is a first attempt.

**Conjecture 11.6** (Hasse Principle for $V(\varphi)$ — C2)**.** For the primitive subfamily $V(\varphi)_{\mathrm{prim}}$ (where $a \times b \neq 0$ and $a, b, c$ are pairwise non-proportional), the Hasse principle holds: $V(\varphi)_{\mathrm{prim}}(\mathbb{Q}) \neq \varnothing$ if and only if $V(\varphi)_{\mathrm{prim}}(\mathbb{Q}_v) \neq \varnothing$ for all places $v$.

*Evidence.* No local-global failures found; the octonionic cross product is surjective modulo every prime tested ($p \leq 13$), so no Brauer-Manin obstruction at finite primes.

**Conjecture 11.7** (No Third Mechanism — S3)**.** Every nontrivial solution to $a^n + b^n = c^n$ in $\mathbb{S}_{\mathbb{Z}}$ (for $n \geq 3$) arises from either Mechanism A (anti-commutativity) or Mechanism B (zero divisors).

**Conjecture 11.8** (Waring Linear Growth — B2)**.** $g(p, \mathbb{H}_{\mathbb{Z}}) \sim p + 1$ for odd primes $p$.

*Evidence.* Three data points: $g(3) \leq 4$, $g(5) \leq 5$, $g(7) \leq 6$.

**Conjecture 11.9** (Zariski Density — C3)**.** $V(\varphi)(\mathbb{Q})$ is Zariski-dense in $V(\varphi)$.

*Argument.* The fiber structure $\varphi(a, b, c) = \langle c, a \times b \rangle$ gives codimension-1 linear fibers in $c$ for each $(a, b)$ with $a \times b \neq 0$. Dense base and dense fibers imply dense total space.

**Conjecture 11.10** ($\varphi$-Contraction as $G_2$ Curvature — F3)**.** The symmetrized contraction identity for the octonionic trilinear form $\varphi$ (proved algebraically in `TrilinearVariety.lean`) encodes the curvature of $G_2$-holonomy manifolds in the sense of Bryant [4] and Joyce [16].

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